

Anticipate biology.

Before it changes the
balance of power.

An evidence-driven research engine
for scientists working on what comes next.



A warning about the way we innovate.

“The United States tends to play catch-up after critical technologies have already become mainstream.”

National Security Commission on Emerging Biotechnology

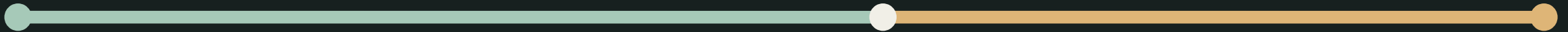
2025 FINAL REPORT / SECTION 1.3

A biological lead can become a unilateral advantage.

Novel biological threats leave less room for catch-up.

Capability develops

Capability becomes visible



ANTICIPATE

REACT

Connect evidence. Test implications.
Prepare defensive options earlier.

Response starts with
less lead time.

TEAM THESIS · Conceptual timeline, not measured intervals. Reactive measures remain necessary.

Point compute at the questions that matter.

Published evidence

- testable connections
- experiments worth reviewing



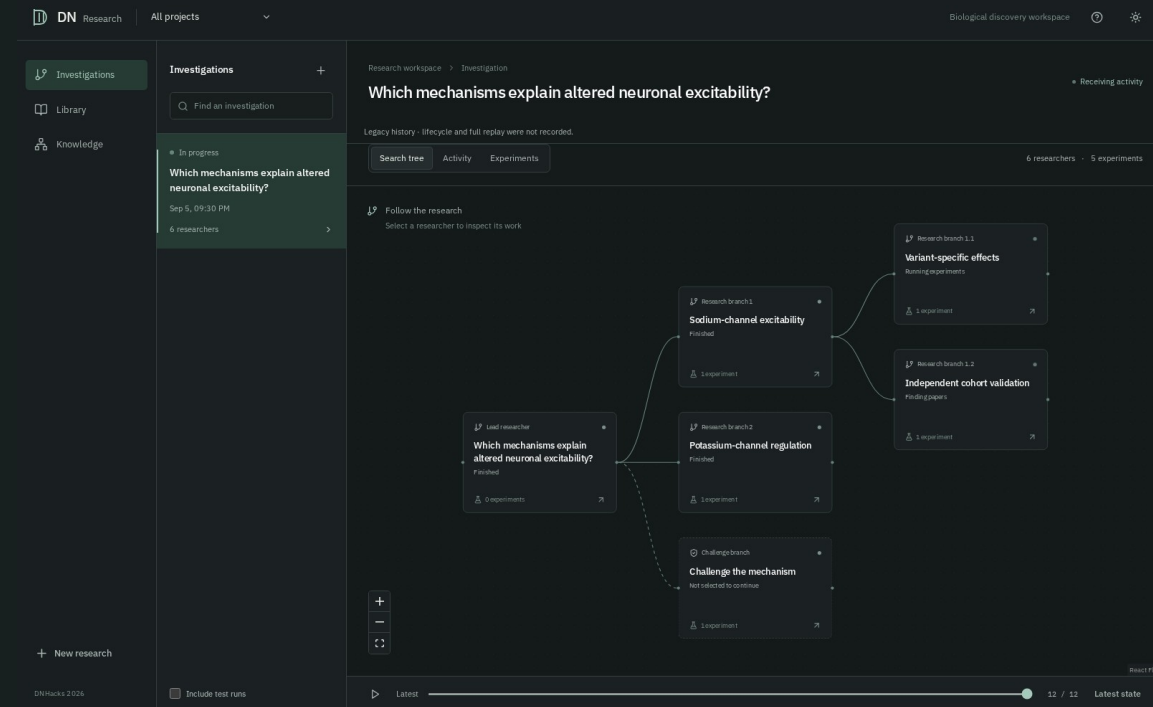
INITIAL PROVING GROUND · Computational biology, including pancreatic-cancer research.

Make the next experiment a deliberate decision.

Computational biologist

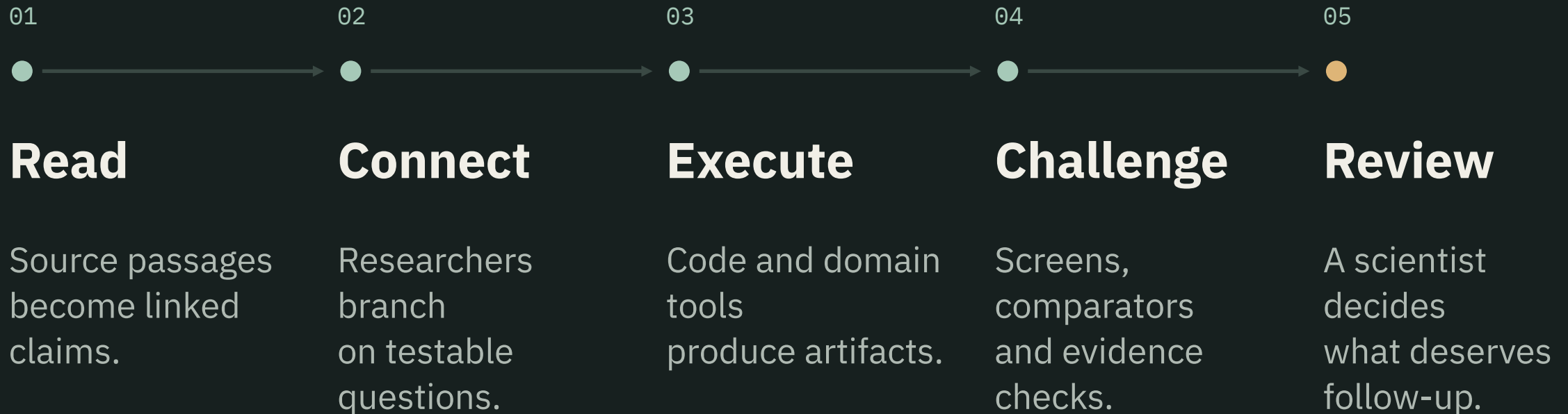
Small research team

Choose a question.
Inspect its evidence.
Explain the decision.



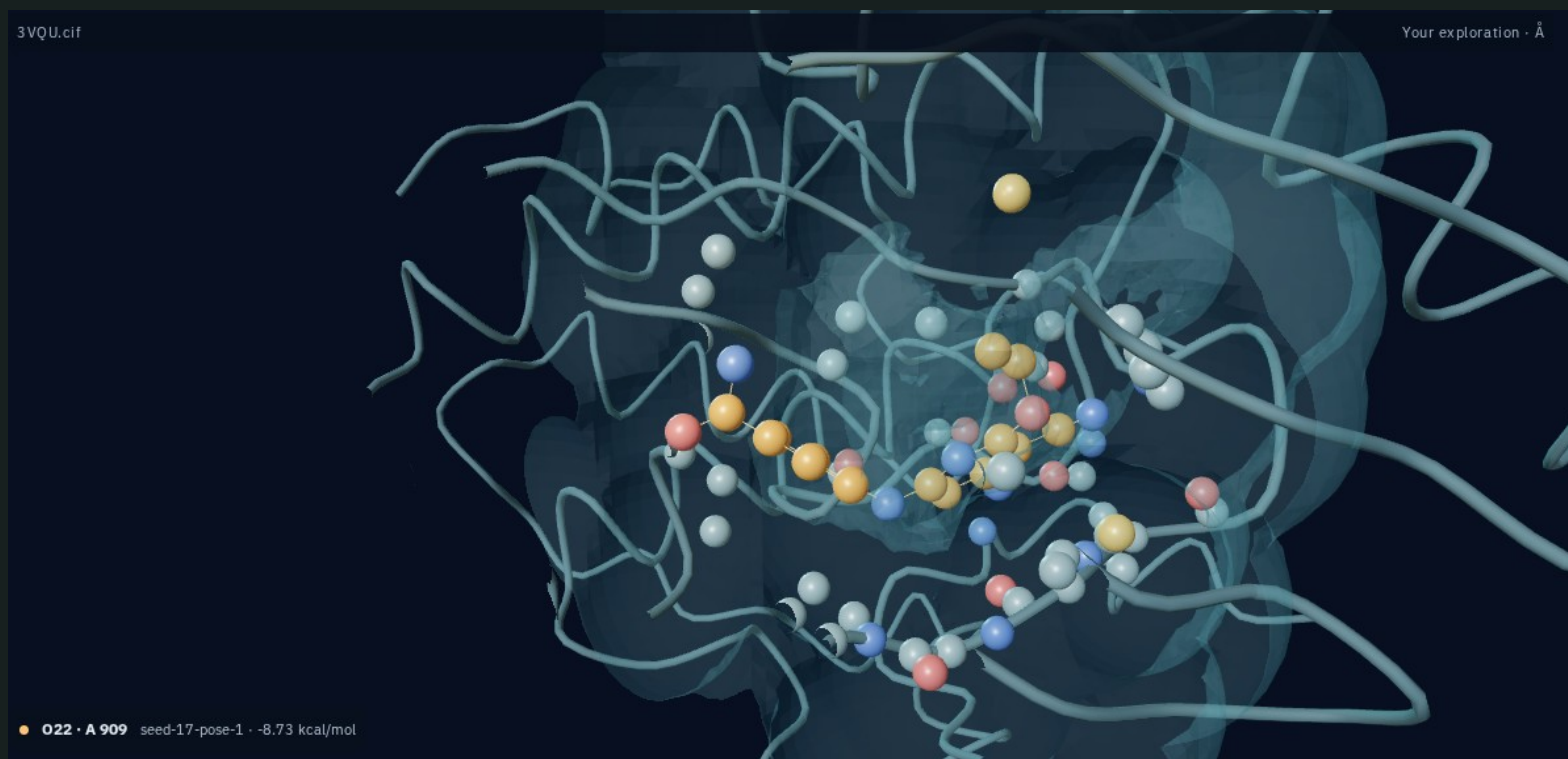
USER HYPOTHESIS · Actual interface with synthetic test data; interviews and usability study remain ahead.

The graph directs exploration. Experiments constrain it.



AI IS FUNDAMENTAL · Question formation, branching, executable analyses and artifact interpretation.

Inspect the image. Check the coordinates.



A real molecular scene

Image critique

Request a cutaway

2.714 Å coordinate check

ACTUAL 3D WORKBENCH · Image-to-coordinate review; the separate docking recovery benchmark did not pass.

Training is real. Advantage still has to be earned.

PROTEIN

219 / 110 / 105

train / validation / development

PCA beats the denoiser
on hidden-value
reconstruction.

DRUG RESPONSE

245 / 40 / 71

train / validation / test groups

PCA selected on validation.
No pancreatic-subset
advantage.

CELL ECOSYSTEMS

45 / 11 / 16

train / validation / external donors

Exact PCA remains best
for external reconstruction.

MEASURED DEVELOPMENT · Different tasks and units. These counts are not pooled or a discovery score.

Evidence needs a budget for being wrong.

INTERNAL E-VALUE TOOLS

1.12%

ever crossed the threshold

112 / 10,000 synthetic null runs
95% Wilson interval: 0.93–1.35%

**Prevent
leakage**

Train on past data; score fresh observations.

**Keep
receipts**

Freeze inputs, consume donors once, preserve replay.

**Respect
limits**

Synthetic calibration is not discovery accuracy.

SEPARATE DIAGNOSTIC · Learned encoder; 96 synthetic pairs/run. No investigation-wide guarantee.

Start with one research team. Earn the right to expand.

Initial buyer	A research lead choosing computational experiments for a small team.
First product	A private research workspace, connected to approved data and tools.
Commercial hypothesis	Workspace subscription plus metered compute; validate willingness to pay.
Pilot acceptance	Useful reviewed hypotheses, independent reruns, total cost and analyst time.

PROPOSED BUSINESS MODEL · No customer, pricing, revenue or cost-saving claims.

The next milestone is proof that survives outside the demo.

REPRODUCE

One bounded investigation

Freeze task and baselines.
Independent scientist rerun.

HARDEN

One private pilot

Separate identities and data.
Constrain access and spend.

VALIDATE

One useful workflow

Measure quality and cost.
Test the user workflow.

RESEARCH-USE ROADMAP · Clinical use, regulated deployment and operational threat forecasting are not validated.

Give scientists a head start.

On the biology that will shape
industrial strength, health and security.

Our ambition: reduce the window
of unilateral biological advantage.

